SEPARABILITY OF A FINITE EXTENSION IS EQUIVALENT TO NONVANISHING OF THE TRACE AND NONDEGENERACY OF THE TRACE PAIRING

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file fields.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: The trace pairing nondegeneracy statement and the discriminant nonvanishing discussion that follows in the source (source lines 2802 sqq and 2831-2836) build on this lemma but are not part of the counted unit.

Editorial note. The label lemma-separable-trace-pairing names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Same chapter: definition-trace-pairing (the symmetric bilinear form Q_{L/K}(alpha, beta) = Trace_{L/K}(alpha beta), source lines 2730-2738), lemma-trace-and-norm-tower (traces compose in towers), lemma-trace-and-norm-from-minimal-polynomial (trace of a power from the minimal polynomial), lemma-separable-first and lemma-finite-purely-inseparable (the purely inseparable degree p layer), lemma-characteristic-vs-minimal-polynomial (minimal = characteristic polynomial for the multiplication map), lemma-recognize-separable (separable iff distinct roots), lemma-sums-of-powers (power sums detect nonvanishing)

Source reference. Original reference 0BIG: sources/stacks-fields/pinned-original-source.tex, source lines 2616–2616. Prerequisite pointer invoked at 2792; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09HI: sources/stacks-fields/pinned-original-source.tex, source lines 1886–1886. Prerequisite pointer invoked at 2775; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09H3: sources/stacks-fields/pinned-original-source.tex, source lines 1424–1424. Prerequisite pointer invoked at 2793; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 030K: sources/stacks-fields/pinned-original-source.tex, source lines 1921–1921. Prerequisite pointer invoked at 2775; it names

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a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0EM9: sources/stacks-fields/pinned-original-source.tex, source lines 1717–1717. Prerequisite pointer invoked at 2799; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0BIH: sources/stacks-fields/pinned-original-source.tex, source lines 2643–2643. Prerequisite pointer invoked at 2767; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0BIJ: sources/stacks-fields/pinned-original-source.tex, source lines 2713–2713. Prerequisite pointer invoked at 2776; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let L/K be a finite extension of fields. The following are equivalent:*

(1) *L/K is separable,*
(2) *$\text{Trace}_{L/K}$ is not identically zero, and*
(3) *the trace pairing $Q_{L/K}$ is nondegenerate.*

Proof. It is clear that (3) implies (2). If (2) holds, then pick $\gamma \in L$ with $\text{Trace}_{L/K}(\gamma) \neq 0$. Then if $\alpha \in L$ is nonzero, we see that $Q_{L/K}(\alpha, \gamma/\alpha) \neq 0$. Hence $Q_{L/K}$ is nondegenerate. This proves the equivalence of (2) and (3).

Suppose that K has characteristic p and $L = K(\alpha)$ with $\alpha \notin K$ and $\alpha^p \in K$. Then $\text{Trace}_{L/K}(1) = p = 0$. For $i = 1, \ldots, p-1$ we see that $x^p - \alpha^{pi}$ is the minimal polynomial for α^i over K and we find $\text{Trace}_{L/K}(\alpha^i) = 0$ by Lemma 0BIH. Hence for this kind of purely inseparable degree p extension we see that $\text{Trace}_{L/K}$ is identically zero.

Assume that L/K is not separable. Then there exists a subfield $L/K'/K$ such that L/K' is a purely inseparable degree p extension as in the previous paragraph, see Lemmas 030K and 09HI. Hence by Lemma 0BIJ we see that $\text{Trace}_{L/K}$ is identically zero.

Assume on the other hand that L/K is separable. By induction on the degree we will show that $\text{Trace}_{L/K}$ is not identically zero. Thus by Lemma 0BIJ we may assume that L/K is generated by a single element α (use that if the trace is nonzero then it is surjective). We have to show that $\text{Trace}_{L/K}(\alpha^e)$ is nonzero for some $e \geq 0$. Let $P = x^d + a_1 x^{d-1} + \ldots + a_d$ be the minimal polynomial of α over K. Then P is also the characteristic polynomial of the linear maps $\alpha : L \to L$, see Lemma 0BIG. Since L/k is separable we see from Lemma 09H3 that P has d pairwise distinct roots $\alpha_1, \ldots, \alpha_d$ in an algebraic closure $\overline{K}$ of K. Thus these are the eigenvalues of $\alpha : L \to L$. By linear algebra, the trace of α^e is equal to $\alpha_1^e + \ldots + \alpha_d^e$. Thus we conclude by Lemma 0EM9. $\square$